State of the Art: Predicting Viral Fitness and Epidemiological Simulation

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1 Introduction

Viral variant surveillance is a global priority for pandemic preparedness, requiring the timely identification of the small fraction of strains that will drive new waves of infection[4]. The SARS-CoV-2 virus has demonstrated rapid mutation, resulting in multiple infection waves[13]. Traditionally, epidemiological models, such as compartmental models, focus on short-term predictions or struggle to incorporate complex interventions and pathogen genetic evolution. This project objective is to build predictive systems that use molecular characteristics to estimate viral fitness and integrate this information into detailed epidemiological simulations.

2 Module 1: Molecular-Level Prediction of Viral Fitness

The first module focuses on developing computational models that predict the transmissibility or epidemic risk of newly emerged SARS-CoV-2 variants based on their molecular and genomic characteristics. This work leverages large-scale genomic surveillance initiatives, with repositories such as GISAID providing millions of sequences for analysis, enabling the development of increasingly sophisticated predictive frameworks.[8, 11, 14, 1] By quantitatively analysing these molecular signals, it becomes possible to identify variants with increased fitness at an early stage, before the widespread circulation.[1] [8]

2.1 Predictive Methodologies and Model Architectures

The landscape of viral fitness prediction has evolved from classical statistical approaches to advanced machine learning architectures that capture complex mutational interactions and evolutionary dynamics.

2.1.1 Statistical Models and Gradient Boosting

Recent state-of-the-art approaches use machine learning models,often gradient boosting algorithms, to estimate the early spread potential of a newly detected variant. These models have achieved high predictive performance, showing that molecular and early epidemiological signals can be informative. The PyR,falta um 0 pequeno, model developed by Obermeyer et al. exemplifies this class, using hierarchical Bayesian multinomial logistic regression on 6.4 million genomes to infer relative fitness of viral lineages. This approach models fitness as a linear combination of individual mutation effects, achieving AUC values approaching $\sim 90\%$ within 1-2 weeks of variant observation. The model successfully identified numerous fitness-enhancing substitutions across the viral proteome, including spike mutations (e.g., N501Y, E484K) and nonspike mutations in nucleocapsid and nonstructural proteins.[11, 2]

However, these linear additive models face fundamental limitations: they cannot account for epistatic interactions between mutations, where the fitness effect of one mutation depends on the genetic background. Additionally, they struggle to predict fitness of variants harboring mutation combinations not yet observed in the training data, limiting their utility for early pandemic warning systems.[2]

2.1.2 Protein Language Models for Fitness Prediction

A major methodological advance involves adapting protein language models (PLMs) transformer based neural networks pretrained on millions of protein sequences—for viral fitness prediction. These models

leverage evolutionary information encoded across vast protein databases to capture context-dependent mutational effects.[7, 5, 9]

CoVFit, developed by Ito et al., represents the first machine learning model designed to directly predict viral fitness rather than relying on statistical inference from population data. Built on ESM-2 (a 650-million parameter transformer model), CoVFit employs a two-stage training strategy: (1) domain adaptation, where ESM-2 is further pretrained on 1,506 Coronaviridae spike protein sequences to specialize in coronavirus biology, creating ESM-2 Coronaviridae ; and (2) multitask finetuning using Low-Rank Adaptation, training on 21,281 genotype-fitness data points (12,817 unique spike protein genotypes across 17 countries) alongside 173,384 deep mutational scanning measurements of antibody escape for 1,548 monoclonal antibodies. This multitask learning framework enables the model to simultaneously predict relative effective reproduction numbers and immune evasion capabilities.[5]

2.2 Molecular Features

Key molecular features used in prediction include:

- **Spike Protein Mutations:** Mutations in the Spike protein are strong predictors of spread risk, although relevant mutations can occur in other genomic regions as well.
- **Genetic Distance (Week-Distance):** Measures such as Jaccard distance quantify how distinct a new variant is from previously dominant ones. Variants that later become dominant typically show larger mutational differences, often reflecting immune escape.
- **Genomic Diversity (Entropy and Heterogeneity):** Entropy-based metrics capture genetic diversity among circulating strains; higher diversity can signal evolutionary dynamics that enable the emergence of fitter variants.

These features act as early indicators rather than deterministic measures of transmissibility, and their predictive value depends on sampling and epidemiological context.

2.3 Model Validation and Performance

Robust validation is essential for distinguishing models that extrapolate to novel variants from those that merely interpolate within training distributions.

2.3.1 Validation Strategies

- **Temporal Splitting:** Training on past data (e.g., pre-2022) and testing on future variants (post-2022) simulates real world deployment.[5]
- **Cross-Geography Generalization:** Training solely data from specific reogions (e.g., UK/USA).[3]
- **Leave One Mutation Out:** Testing each mutation's effect in isolation while training on all others.[5]
- **Ablation Studies:** Removing components (e.g., DMS data, domain adaptation) to quantify their contribution.[5]

3 Module 2: Epidemiological Simulation and Spread

To capture the heterogeneous dynamics of variant spread, this project employs Agent-Based Models (ABMs), which offer finer granularity and flexibility than traditional compartmental models (e.g., SIR). ABMs explicitly represent individual agents and their interactions, enabling the simulation of population heterogeneity, complex interventions, and realistic contact structures.

3.1 Agent-Based Modeling Frameworks

State-of-the-art ABM platforms such as Covasim [6, 12] and PhASETraCE [10] model transmission dynamics by simulating each individual as an autonomous agent characterized by demographic attributes (e.g., age, comorbidities), disease states (susceptible, exposed, infectious, recovered, deceased), and realistic contact networks (e.g., households, schools, workplaces, community settings). By incorporating these features, ABMs can reproduce population-level patterns of infection spread, intervention effects, and behavioral heterogeneity.

3.2 Dataset Selection for Epidemiological Simulation

For the epidemiological simulation in Module 2, this project uses data from the Our World in Data (OWID) repository, one of the most widely used sources in academic studies and official pandemic reports. It should be noted that the GitHub datasets used in this project are no longer updated after 19 August 2024, and therefore all analyses are based on data available up to that date. These data are open-access, reproducible, and regularly updated, ensuring scientific credibility and thereby facilitating the reproducibility of the simulation work.

OWID provides extensive time-series data spanning the entire COVID-19 pandemic. The repository provides information on key epidemiological variables, including:

- **Confirmed cases**
- **Hospitalizations**
- **Mortality**
- **Vaccination rates**
- **Indirect indicators of intervention measures**

While Module 1 produces molecular-level fitness estimates, OWID data provides context and empirical evidence for testing molecular hypotheses.

4 Conclusion

The current State of the Art shows that combining molecular characteristics with epidemiological simulation provides a promising framework for anticipating the emergence and spread of new variants. Module 1 demonstrates that large-scale genomic surveillance and machine learning can detect early molecular signatures linked to viral fitness, while Module 2 highlights how agent-based models capture realistic transmission dynamics and intervention effects. Together, these approaches illustrate a shift toward multi-scale systems that connect viral evolution with population-level outcomes, offering a more adaptive foundation for future infectious disease surveillance.

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